

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M. Sadhara ¹⁹²⁵²⁴³⁸⁶ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Cryptography and Networks

Reg no: 192524386

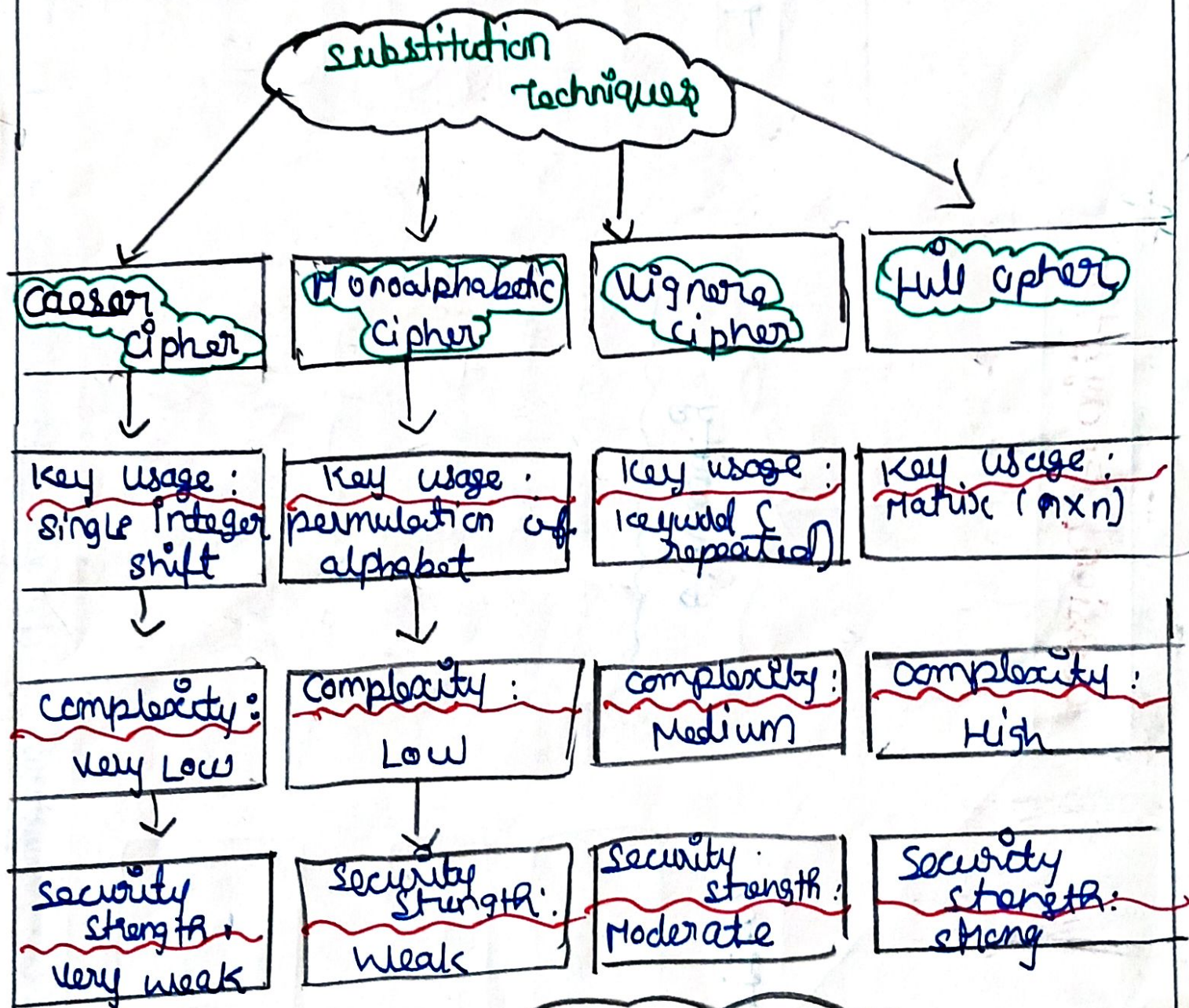
name: M. Sadhana

Course
Code: CSA5116

Date: 06/10/2026.

ASSIGNMENT-2

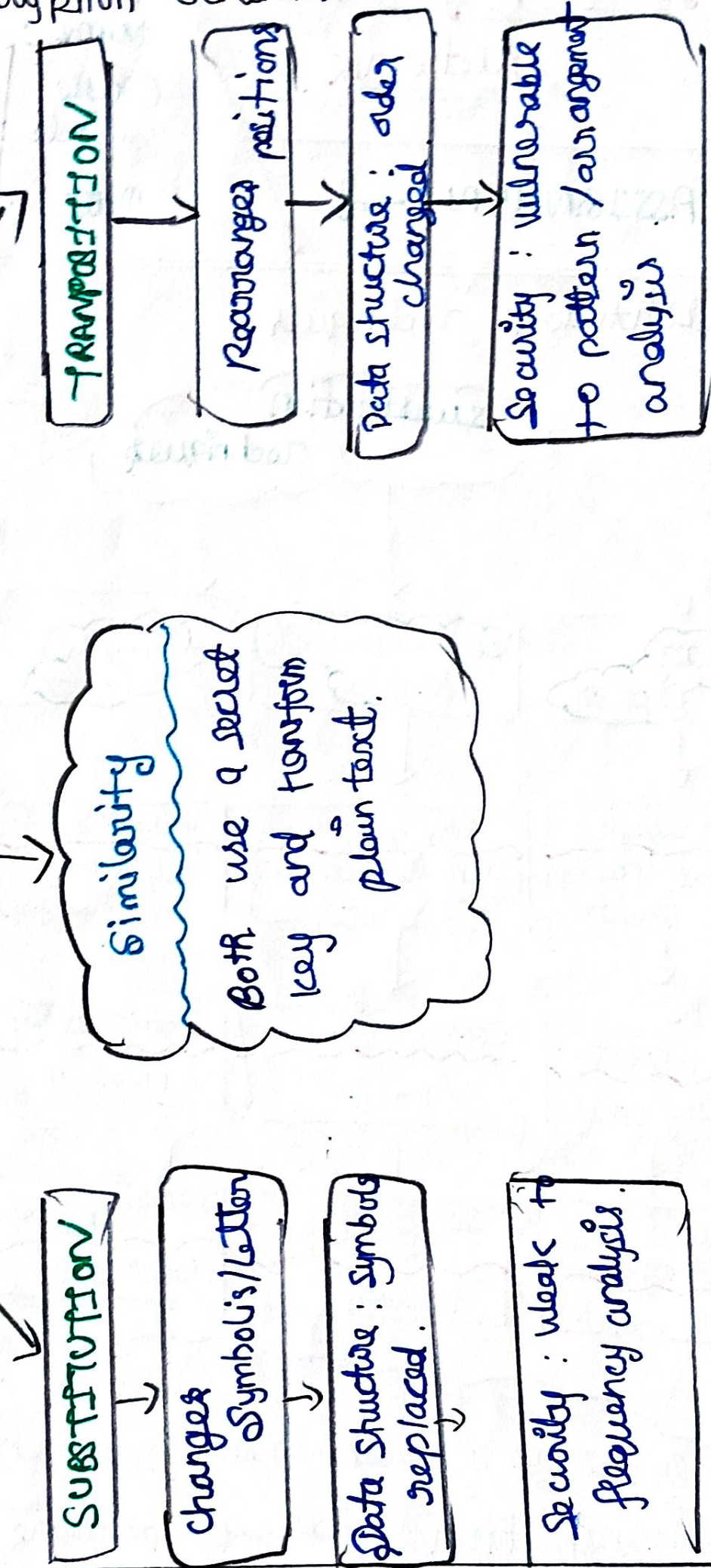
2). Substitution Techniques:



Difference: substitution changes characters =
Transposition changes positions.

3. Classical Encryption Schemes

classical Encryption



Difference: Substitution changes characters; Transposition changes positions.

Classical Encryption Techniques:

classical encryption is used to protect information by converting the original message (plain text) into an unreadable form called cipher text. the receiver uses a suitable method or key to recover the original message.

1. Substitution technique:

In substitution technique, the original letters are replaced with other letter or symbols.

Monalphabetic cipher:

Each letter is replaced by a fixed different letter.

eg:

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

- the same substitution is used throughout the message.

- It is a simple but can be attacked using frequency analysis.

Transposition Techniques:

In transposition techniques, the letters are not replaced. Instead, their positions are replaced.

- The original letters remain the same.
- Only their order is changed.
- Example: HELLO $\rightarrow$ LHOEL
- It provides better security when combined with other techniques.

3. Cryptanalysis.

Cryptanalysis means analyzing an encrypted message.

- It studies the ciphertext.
- It tries to find the plaintext or key.
- Different techniques such as frequency analysis can be used.

Security:
The main purpose of encryption is to provide confidentiality and security.